

LENANA SCHOOL 2025 PREDICTION



POSSIBLE KCSE EXAM CHEMISTRY



233/2

PAPER 2

TIME: 2 HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES

- (a) Write your name and class number in each of the foolscaps answer sheets.*
- (b) Sign and write the date of examination in each of the foolscap answer sheets.*
- (c) Answer **all** the questions in this paper on one side of each foolscap.*

FOR EXAMINER'S USE ONLY

Question Number	Maximum score	Candidate's score
1	13	
2	12	
3	9	
4	10	
5	11	
6	12	
7	13	
TOTAL	80	

Answer **all** the questions in this paper

1. The diagram shows part of the Periodic Table. The letters are not the actual symbols of the elements.

[illegible]

(a) Compare the reactivity between elements Q and T. **(2 marks)**

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(b) Explain the electrical conductivity of the chloride of element T. **(2 marks)**

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(c) Compare the melting and boiling points of elements R and S. (2 marks)

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(d) Write an equation for the second ionization energy of element Q. **(1mark)**

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(e) How does the atomic and ionic radius of each of the following elements compare?

(i) Element T. (1½ marks)

(ii)Element P (1½ marks)

(f) Compare atomic radius of elements R and Q. (1½ marks)

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(g) Explain the difference in the melting points of the oxides of element Q and the oxide of element R. (1½ marks)

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2 (a) The diagram below shows the structure of an allotrope of sulphur



(i) What are allotropes?

(1mark)

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(ii) Identify the allotrope shown in the diagram above.

(1mark)

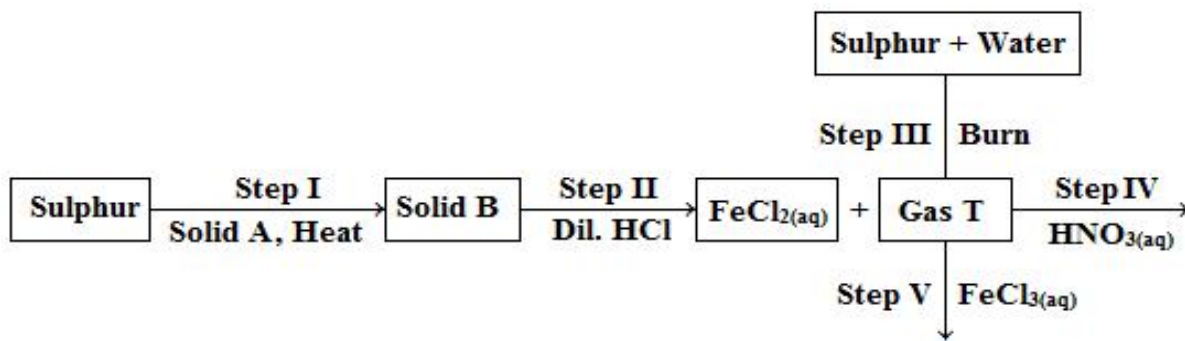
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(iii) State **two** properties of the allotrope above.

(2 marks)

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(b) Study the flow chart below and answer the questions that follow.



(i) Write the equation for the reactions in:

I. step I.

(1mark)

II. Step II.

(1mark)

(ii) State **two** observations made in step II.

(2 marks)

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(iii) Explain the observations made in:

I. Step IV.

(1½ marks)

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II. Step V.

(1½ marks)

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(iv) State **one** use of gas T.

(1mark)

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3. Next to each letter, A to F, in the table below is the molecular formula of an organic compound.

A	C ₂ H ₅ Br	B	C ₂ H ₄
C	C ₄ H ₁₀	D	C ₂ H ₆ O
E	C ₃ H ₆ O	F	C ₃ H ₆ O ₂

(a) Choose a molecular formula above that represents an organic compound below. Write down only the letter (A to F) next to the question numbers

(i) A haloalkane

(½ mark)

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(ii) An alcohol **(½ mark)**

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(iii) An unsaturated hydrocarbon **(½ mark)**

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(iv) A product of thermal cracking of compound C. **(½ mark)**

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(b) If compound F is a carboxylic acid, write down the following:

(i) The structural formula of a functional isomer (an isomer with a different functional group) of F.

(1mark)

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(ii) The IUPAC name of a functional isomer of F. **(1mark)**

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(c) Compound B is a monomer used to make a polymer. Write down the:

(i) Definition of a polymer. **(1mark)**

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(ii) IUPAC name of the polymer. **(1mark)**

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(iii) Balanced equation for the polymerisation reaction **(1mark)**

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(d) Compound A is used as a reactant in the production of compound D.

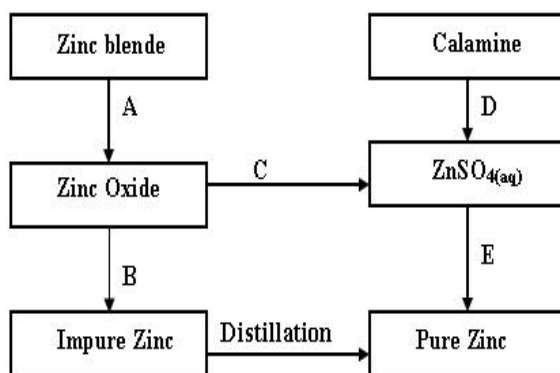
(i) Name the type of reaction that takes place. (1mark)

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(ii) State two changes that can be made to the reaction conditions in (d) (i) to obtain compound B, instead of D, as product. (2 marks)

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4. The flow chart below summarizes the extraction of Zinc, study it and answer the questions that follow.



(a) Name the process represented by A and B

A.....

(2 marks)

B

(b) Identify the reagents required for process B, C and D

(3 marks)

B

C

D

(c) Write a chemical equation of the reaction that occurs in process B

(1mark)

(d) With an aid of a diagram, explain how you would obtain a pure sample of Zinc by process E.
(2 marks)

(e) State two uses of Zinc metal **(2 marks)**

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5 (a) The table below gives some values of standard enthalpies of formation ($\Delta H_f^\ominus$).

Substance	$F_{2(g)}$	$CF_{4(g)}$	$HF_{(g)}$
$\Delta H_f^\ominus$ (kJmole ⁻¹)	0	– 680	– 269

The enthalpy change for the reaction $C_2H_{6(g)} + 7F_{2(g)} \rightarrow 2CF_{4(g)} + 6HF_{(g)}$ is $-2889 \text{ kJ mol}^{-1}$.

Use this value and the standard enthalpies of formation in **Table 2** to calculate the standard enthalpy of formation of $C_2H_{6(g)}$. **(3 marks)**

(b) In an experiment to determine the enthalpy of solution of concentrated sulphuric (VI) acid (specific gravity = 1.84gcm^{-3}) the following procedure was used:

- A clean 250.0 cm^3 glass or plastic beaker is wrapped with a newspaper leaf.
- About 50.0 cm^3 of tap water is measured into the beaker and the steady temperature noted.
- The beaker is held in a tilted position and 2.0 cm^3 of and sulphuric acid added into the water

(i) Why was the beaker wrapped with newspaper leaf? **(1mark)**

(ii) Why was the acid added into water and not vice versa? **(1mark)**

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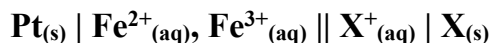
(iii) **Explain** the reason for tilting the beaker during addition of the acid into water. **(2 marks)**

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(iv) Calculate the molar enthalpy of solution of concentrated sulphuric (VI) acid given that ΔT for the reaction = 1°C . (Density of water = 1gcm^{-3} ; specific heat capacity of water = $4.2\text{kJkg}^{-1}\text{K}^{-1}$).

(4 marks)

6 (a) Consider the electrochemical cell represented by the cell notation below, where X is an unknown metal:



(The cell potential of this cell was found to be 0.03 V.

(i) Write down the type of electrochemical cell illustrated above. **(1mark)**

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(ii) What does the single line (|) in the above cell notation represent? **(1mark)**

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(iii) Write down the half-reaction that takes place at the anode in the above cell. **(1mark)**

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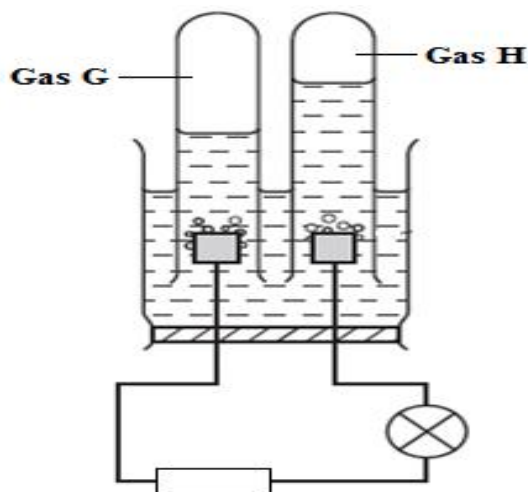
(iv) Given that:

Half reaction	$E^{\ominus}$ (volts)
$\text{Fe}^{3+}_{(\text{aq})} + \text{e} \rightarrow \text{Fe}^{2+}_{(\text{aq})}$	+ 0.77
$\text{Ag}^{+}_{(\text{aq})} + \text{e} \rightarrow \text{Ag}_{(\text{s})}$	+ 0.80
$\text{Na}^{+}_{(\text{aq})} + \text{e} \rightarrow \text{Na}_{(\text{s})}$	− 2.87
$\text{K}^{+}_{(\text{aq})} + \text{e} \rightarrow \text{K}_{(\text{s})}$	− 2.92

Identify X with the aid of a calculation.

(2 marks)

- (b)** The diagram below shows the apparatus that can be used to electrolyse dilute Sulphuric acid. Study it to answer the questions that follow.



- (i)** Identify the gases H and G

I. H (1mark)

II. G (1mark)

- (ii)** What happens to the concentration of the Sulphuric acid during the process with time? Explain
(2 marks)

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- (iii)** During the electrolysis a current of 0.72A was passed through the electrolyte for 15 minutes.
Calculate the volume of gas H produced. (1 Faraday = 96,500C, molar gas volume = 24 dm³ at r.t.p).
(3 marks)

7 (a) (i) What is a salt?

(1mark)

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(ii) Write the formula of any **two** double salts.

(1mark)

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(b) A student has found that her sample of potassium nitrate is contaminated with small amounts of a green solid. She picks out a small piece of the green solid and finds that it is insoluble in water.

(i) Describe how you would make a pure sample of potassium nitrate from the impure mixture.

(3marks)

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(ii) The student believes that the green solid is copper (II) carbonate. Describe a series of 3 tests that the student could use to confirm this.

(6 marks)

Test	Procedure	Observations	Conclusion
1			
2			

3			

- (c) In an experiment 50g of a saturated solution of a salt X was heated to dryness in an evaporating dish.
The mass of crystals when weighed gave a reading of 2.7g. Determine the solubility of the salt.

(2 marks)

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